

Part 5 Payout Policy and Capital Structure

CHAPTER

16

Does Debt Policy Matter?

A corporation's mix of debt and equity financing is called its *capital structure*. When the corporation uses debt, it is said to employ *financial leverage* or *gearing*. What happens if the financial manager uses more or less financial leverage? Is there an ideal mix of debt and equity financing that maximizes the market value of the firm?

There is a traditional view—still lurking in some corners of finance—that debt financing adds value just because the interest rate paid to debt holders is less than the cost of equity. But Modigliani and Miller (MM) proved that market value does *not* depend on capital structure in perfect, frictionless financial markets. MM's famous Proposition 1 states that the total value of outstanding debt and equity depends on the value of the firm's real assets, operations, and growth opportunities, but not on the proportions of debt and equity.

MM's Proposition 1 allows complete separation of investment and financing decisions. It implies that any firm could use the capital budgeting procedures presented in Chapters 5 through 11 without worrying about where the money for capital expenditures comes from. In those chapters, we assumed all-equity financing without really thinking about it. If MM are correct, that is exactly the right approach. If the firm uses a mix of debt and equity financing, its overall cost of capital will be exactly the same as if it were financed entirely with equity.

Financing decisions do matter in practice, for reasons detailed in Chapters 17 and 18. But we devote this chapter to MM because their proposition is the starting point for all applied capital structure theory. If you don't understand the starting point, you won't understand the destination.

Section 16-1 Financial leverage and shareholder value

Section 16-1 introduces capital-structure decisions and explains when maximizing the *total* market value of the firm also maximizes shareholder value.

Section 16-2 Modigliani and Miller's Proposition 1

Section 16-2 sets out MM's Proposition 1 that capital structure is irrelevant to value. We provide a simple numerical example to illustrate MM's argument.

Section 16-3 Leverage and expected returns: MM's Proposition 2

Financial leverage creates *financial risk*. When a corporation finances with more debt and less equity, the risk of the remaining equity goes up and its cost goes up to compensate. MM showed that the increased cost of equity exactly offsets the advantage of borrowing at interest rates lower than the cost of equity.

Section 16-4 No magic in financial leverage

Section 16-4 considers whether some special types of corporate securities could have extra value to investors, thus undercutting MM's logic. We can't rule out this possibility, but, if managers spot an opportunity to issue such securities, there will be an increase in supply that eliminates any gains from further issues of these. This kind of supply response to meet demand should reassure financial managers that MM's proposition is at least approximately correct most of the time.

Section 16-5 A final word on the cost of capital

Throughout this chapter we ignore corporate income taxes, except in Section 16-5, where we recognize the tax deductibility of interest expense on corporate borrowing. We show how this interest tax shield is accounted for in the *after-tax weighted average cost of capital* (WACC).

16-1 Financial Leverage and Shareholder Value

Financial managers try to find the combination of debt and equity financing that maximizes the total value of the firm, debt and equity combined. Before tackling this problem, we should confirm that the financing policy that maximizes total value also maximizes shareholder value.

Example A simplified example will suffice. Let D and E represent the market values of the debt and equity of the Wapshot Wind Farms Company. Wapshot's 1,000 shares sell for \$50 apiece. Thus, Wapshot's equity is

$$E = 1,000 \times 50 = \$50,000$$

Wapshot has also borrowed \$25,000, and so V , the total market value of all Wapshot's outstanding securities, is

$$V = D + E = \$75,000$$

Wapshot's stock is known as *levered equity*. Its stockholders face the benefits and costs of **financial leverage**, or *gearing*. Suppose that Wapshot "levers up" still further by borrowing an additional \$10,000 and paying the proceeds to shareholders as a special dividend of \$10 per share. This substitutes debt for equity capital with no impact on Wapshot's assets.

What will Wapshot’s equity be worth after the special dividend is paid? We have two unknowns, E and V :

Old debt	\$25,000	} \$35,000 = D
New debt	\$10,000	
Equity		? = E
Firm value		? = V

If V is \$75,000 as before, then E must be $V - D = 75,000 - 35,000 = \$40,000$. Stockholders have suffered a capital loss that exactly offsets the \$10,000 special dividend. But if V increases to, say, \$80,000 as a result of the change in capital structure, then $E = \$45,000$ and the stockholders are \$5,000 ahead. In general, because the new debtholders can demand only a fair rate of return, any increase or decrease in V caused by a shift in capital structure accrues to the firm’s stockholders. We conclude that *the policy that maximizes the market value of the firm is also best for the firm’s stockholders.*

16.1 Self-Test

Ventone invests in a new project with an NPV of \$5 million. It partly finances the venture by borrowing \$3 million. The remaining investment of \$10 million is financed out of retained earnings. Does the value of the equity increase by \$5 million, or by less? Do the new debtholders get a share of the NPV?

This conclusion rests on two important assumptions: First, that Wapshot’s shareholders do not gain or lose from payout policy and, second, that after the change in capital structure the old and new debt are together *worth* \$35,000.

Payout policy may or may not be relevant, but there is no need to repeat the discussion of Chapter 15. We need only note that shifts in capital structure sometimes force important decisions about payout policy. Perhaps Wapshot’s cash dividend has costs or benefits that should be considered in addition to the benefits, if any, achieved by its increased financial leverage.

Our second assumption that old plus new debt ends up worth \$35,000 seems unimportant. But it could be wrong. Perhaps the new borrowing has increased the risk of the old bonds. If the holders of old bonds cannot demand a higher rate of interest to compensate for the increased risk, the value of their investment is reduced. In this case, Wapshot’s stockholders gain at the expense of the holders of old bonds, even though the overall value of the firm is unchanged.

But this anticipates issues better left to Chapter 17. In this chapter, we assume that any

new issue of debt has no effect on the market value of existing debt.

16-2

Modigliani and Miller's Proposition 1

Let us accept that the financial manager would like to find the capital structure that maximizes the value of the firm. How is this done? MM's answer is that the financial manager should stop worrying: In a perfect market, any financing package is as good as another. The value of the firm is unaffected by its choice of capital structure.¹ That is not such an odd statement as it may initially sound. In Chapter 12, we pointed out that when newly issued securities are fairly priced, the issue has zero NPV. In that case, you would not expect a financial manager to be able to add value by substituting one zero-NPV security for another. For example, suppose that a company changes its capital structure by issuing bonds. The investors who buy those bonds must have calculated that they offered fair value and therefore had a zero NPV. Issuing a bond at zero NPV cannot enhance the company's value.

We need to look more closely at why this is the case. Imagine two firms that generate the same stream of operating profits and differ only in their capital structure. Firm U is unlevered. Therefore, the total value of its equity E_U is the same as the total value of the firm V_U : $E_U = V_U$. Firm L, on the other hand, is levered. The value of its equity is, therefore, equal to the value of the firm less the value of the debt: $E_L = V_L - D_L$.

Now think which of these firms you would prefer to invest in. If you don't want to take much risk, you can buy common stock in the unlevered firm U. For example, if you buy 1% of firm U's shares, your investment is $0.01 V_U$ and you are entitled to 1% of the operating profits:

Dollar Investment	Dollar Return
$0.01 V_U$	$0.01 \times \text{Profits}$

Now compare this with an alternative strategy. This is to purchase the same fraction of *both* the debt and the equity of firm L. Your investment and return are then:

	Dollar Investment	Dollar Return
Debt	$0.01 D_L$	$0.01 \times \text{Interest}$
Equity	$0.01 E_L$	$0.01 \times (\text{Profit} - \text{Interest})$
Total	$0.01(D_L + E_L)$ $= 0.01 V_L$	$0.01 \times \text{Profits}$

Both strategies offer the same payoff: 1% of the firm's profits. The law of one price tells us that in well-functioning markets two investments that offer the same payoff must have the same price. Therefore, $0.01 V_U$ must equal $0.01 V_L$: The value of the unlevered firm must equal the value of the levered firm.

Suppose that you are willing to run a little more risk. You decide to buy 1% of the outstanding shares in the *levered* firm. Your investment and return are now:

Dollar Investment	Dollar Return
$0.01 E_L$ $= 0.01(V_L - D_L)$	$0.01 \times (\text{Profit} - \text{Interest})$

Again, there is an alternative strategy. This is to borrow $0.01 D_L$ on your own account and purchase 1% of the stock of the unlevered firm.² In this case, your strategy gives you 1% of the profits from V_U , but you have to pay interest on your loan equal to 1% of the interest that is paid by firm L. Your total investment and net return are:

	Dollar Investment	Dollar Return
Borrowing	$-0.01 D_L$	$-0.01 \times \text{Interest}$
Equity	$0.01 V_U$	$0.01 \times \text{Profits}$
Total	$0.01(V_U - D_L)$	$0.01 \times (\text{Profits} - \text{Interest})$

Once more, both strategies offer the same payoff: 1% of profits after interest. Therefore, both investments must have the same cost. The investment $0.01(V_U - D_L)$ must equal $0.01(V_L - D_L)$ and V_U must equal V_L .

This picture of individual investors borrowing money to buy shares in firm U may seem unrealistic. If so, think instead of a large institutional investor—for example, a pension fund. Such a fund rarely borrows money. It is a lender because it invests in bonds as well as stocks. Suppose the fund wants to buy 1% of the stock of firm U, but “lever up” so that its investment outlay equals only the cost of buying 1% of the stock of firm L. Instead of borrowing, it sells bonds equal to 1% of the debt of firm L and uses the proceeds of the sale to pay for part of its investment in U's equity. By doing so, it loses bond interest equal to what it would pay to borrow 1% of D_L . The result is:

	Dollar Investment	Dollar Return
Sale of bonds	$-0.01 D_L$	$-0.01 \times \text{Interest foregone}$
Equity	$0.01 V_U$	$0.01 \times \text{Profits}$
Total	$0.01(V_U - D_L)$	$0.01 \times (\text{Profits} - \text{Interest})$

Again we have the MM result. The payoffs from buying firm U's stock, partly financed by selling some bonds, are the same as the payoffs from buying L's stock (and also the same as

buying U's stock with borrowed money). U and L must therefore be worth the same overall.

As long as investors can borrow or lend on the same terms as the firm, they can “undo” the effect of any changes in the firm's capital structure. This is how MM arrived at their famous Proposition 1: *The market value of any firm is independent of its capital structure.*

The Law of the Conservation of Value

MM's argument that debt policy is irrelevant is an application of an astonishingly simple idea. If we have two streams of cash flow, *A* and *B*, then the present value of $A + B$ is equal to the present value of *A* plus the present value of *B*. That's common sense: If you Page 463
have a dollar in your left pocket and a dollar in your right, your total wealth is \$2.

We met this principle of *value additivity* in our discussion of capital budgeting, where we saw that the present value of two assets combined is equal to the sum of their present values considered separately.

In the present context, we are not combining assets but splitting them up. But value additivity works just as well in reverse. We can slice a cash flow into as many parts as we like; the values of the parts will always sum back to the value of the unsliced stream. (Of course, we have to make sure that none of the stream is lost in the slicing. We cannot say, “The value of a pie is independent of how it is sliced,” if the slicer is also a nibbler.)

This is really a *law of the conservation of value*. The value of an asset is preserved regardless of the nature of the claims against it. Thus, Proposition 1: Firm value is determined on the *left-hand* side of the balance sheet by real assets—not by the proportions of debt and equity securities issued to buy the assets.

The simplest ideas often have the widest application. For example, we could apply the law of the conservation of value to the choice between raising \$100 million by issuing preferred stock, common stock, or some combination. The law implies that the choice is irrelevant, assuming perfect capital markets and providing that the choice does not affect the firm's investment and operating policies. If the total value of the equity “pie” (preferred and common combined) is fixed, the firm's owners (its common stockholders) do not care how this equity pie is sliced.

The law also applies to the mix of debt securities issued by the firm. The choices of long-term versus short-term, secured versus unsecured, senior versus subordinated, and convertible versus nonconvertible debt all should have no effect on the overall value of the firm.

Combining assets and splitting them up will not affect values as long as they do not affect investors' choices. When we showed that capital structure does not affect choice, we implicitly assumed that both companies and investors can borrow and lend at the same risk-free rate of interest. As long as this is so, investors can undo the effect of any changes in the firm's capital structure.

16.2 Self-Test

Endo Horrocks has operating income of \$12.5 million. It is financed solely by equity that has a market value of \$100 million. Suppose that the company now issues \$50 million of debt at an interest rate of 10% and uses the cash to pay a special dividend to stockholders. The pie chart below show the operating income and the company value before the debt issue. Draw two similar pies to show how MM would predict the income and company value to be divided after the debt issue.



An Example of Proposition 1

Macbeth Spot Removers is reviewing its capital structure. Table 16.1 shows its current position. The company has no leverage, and all the operating income is paid as dividends to the common stockholders (we assume still that there are no taxes). The expected earnings and dividends per share are \$1.50, but this figure is by no means certain—it could turn out to be more or less than \$1.50. The price of each share is \$10. Because the firm expects to produce a level stream of earnings in perpetuity, the expected return on the share is equal to the earnings-price ratio, $1.50/10.00 = 0.15$, or 15%.

Macbeth Spot Removers				
Number of shares	1,000			
Price per share	\$ 10			
Market value of shares	\$10,000			
Outcomes				
Operating income (\$)	500	1,000	1,500	2,000
Earnings per share (\$)	0.50	1.00	1.50	2.00
Return on shares (%)	5	10	15	20
			Expected outcome	

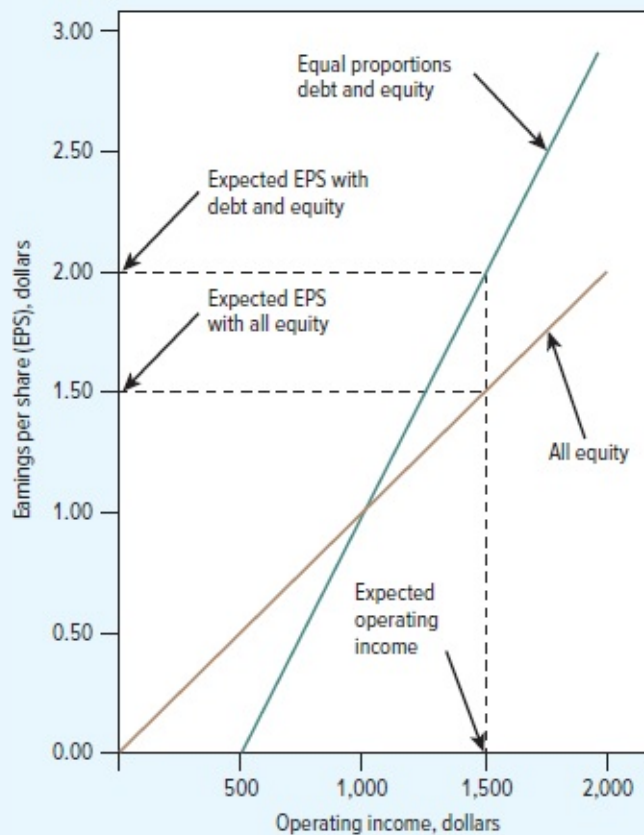
TABLE 16.1 Macbeth Spot Removers is entirely equity financed. Although it expects to have an income of \$1,500 a year in perpetuity, this income is not certain. This table shows the return to the shareholder under different assumptions about operating income. We assume no taxes.

Ms. Macbeth, the firm's president, has concluded that shareholders would be better off if the company had equal proportions of debt and equity. She therefore proposes to issue \$5,000 of debt at an interest rate of 10% and to use the proceeds to repurchase 500 shares. To support her proposal, Ms. Macbeth has analyzed the situation under different assumptions about operating income. The results of her calculations are shown in Table 16.2.

Possible Payoffs to Equity				
Number of shares	500			
Price per share	\$ 10			
Market value of shares	\$5,000			
Market value of debt	\$5,000			
Interest at 10%	\$ 500			
Outcomes				
Operating income (\$)	500	1,000	1,500	2,000
Interest (\$)	500	500	500	500
Equity earnings (\$)	0	500	1,000	1,500
Earnings per share (\$)	0	1.00	2.00	3.00
Return on shares (%)	0	10	20	30
				Expected outcome

TABLE 16.2 Macbeth Spot Removers is wondering whether to issue \$5,000 of debt at an interest rate of 10% and repurchase 500 shares. This table shows the return to the shareholder under different assumptions about operating income.

To illustrate how leverage would affect earnings per share, Ms. Macbeth has also produced Figure 16.1. The brown line shows how earnings per share would vary with operating income under the firm's current all-equity financing. It is, therefore, simply a plot of the data for the four outcomes of operating income and earnings per share in Table 16.1. The green line shows how earnings per share would vary with operating income given equal proportions of debt and equity. It is, therefore, a plot of the data in Table 16.2.



► **FIGURE 16.1**

Borrowing increases Macbeth's EPS (earnings per share) when operating income is greater than \$1,000 and reduces EPS when operating income is less than \$1,000. Expected EPS rises from \$1.50 to \$2.

Ms. Macbeth reasons as follows: “It is clear that the effect of leverage depends on the company's income. If income is greater than \$1,000, the return to the shareholder is *increased* by leverage. If it is less than \$1,000, the return is *reduced* by leverage. The return is unaffected when operating income is exactly \$1,000. At this point the return on the market value of the assets is 10%, which is exactly equal to the interest rate on the debt. Our capital structure decision, therefore, boils down to what we think about the company's prospects. Since we expect operating income to be above the \$1,000 break-even point, I believe we can best help our shareholders by going ahead with the \$5,000 debt issue.”

As financial manager of Macbeth Spot Removers, you reply: “Your argument ignores the fact that investors could borrow on their own account. For example, suppose that an investor puts up \$10 of his own money, borrows a further \$10, and then invests the total in two unlevered Macbeth shares. This is exactly the same set of payoffs as the investor would get by buying one share in the levered company. [Compare the last two lines of Tables 16.2 and 16.3.] Therefore, a share in the levered company must also sell for \$10. If Macbeth goes ahead and borrows, it will not allow investors to do anything that they could not do already,

and so it will not increase value.”

Payoffs from borrowing to buy Macbeth shares				
	Outcomes			
Earnings on two shares(\$)	1.00	2.00	3.00	4.00
Less interest at 10% (\$)	1.00	1.00	1.00	1.00
Net earnings on investment (\$)	0	1.00	2.00	3.00
Return on \$10 investment (%)	0	10	20	30
			Expected outcome	

TABLE 16.3 Any investor can replicate Macbeth’s leverage.

The argument that you are using is exactly the same as the one MM used to prove Proposition 1.

16-3

Leverage and Expected Returns:
MM’s Proposition 2

When MM first proposed Proposition 1, it seemed a radical suggestion. Economists and financial managers were used to the idea that “cheap” debt could enhance earnings per share. But the *value* of a share depends not only on the payments to shareholders, but also on the rate at which those cash flows are discounted.

Consider the implications of MM’s Proposition 1 for the expected returns on Macbeth stock:

	Current Structure: All Equity	Proposed Structure: Equal Debt and Equity
Expected earnings per share (\$)	1.50	2.00
Price per share (\$)	10	10

Leverage increases the expected stream of earnings per share but *not* the share price. The reason is that the change in the expected earnings stream is exactly offset by a change in the rate at which the earnings are discounted. The expected return on the share (which for a perpetuity is equal to the earnings–price ratio) increases from 15% to 20%. We now show how this comes about.

The expected return on Macbeth's assets r_A is equal to the expected operating income calculated before interest expense, divided by the total market value of the firm's securities:

$$\text{Expected return on assets} = r_A = \frac{\text{expected operating income}}{\text{market value of all securities}}$$

We have seen that in perfect capital markets the company's borrowing decision does not affect *either* the firm's operating income *or* the total market value of its securities. Therefore, the borrowing decision also does not affect the expected return on the firm's assets r_A .

Suppose that an investor holds all of a company's debt and all of its equity. This investor is entitled to all the firm's operating income; therefore, the expected return on the portfolio is just r_A .

The expected return on a portfolio is equal to a weighted average of the expected returns on the individual holdings. Therefore, the expected return on a portfolio consisting of *all* the firm's securities is

$$\begin{aligned} \text{Expected return on assets} &= (\text{proportion in debt} \times \text{expected return on debt}) \\ &\quad + (\text{proportion in equity} \times \text{expected return on equity}) \\ r_A &= \left(\frac{D}{D+E} \times r_D \right) + \left(\frac{E}{D+E} \times r_E \right) \end{aligned} \tag{16.1}$$

This formula is, of course, an old friend from Chapter 9. The expected returns on debt and equity are called the *cost of debt* and the *cost of equity*. The overall expected return r_A is called the *company cost of capital*. MM's Proposition 1 says that r_A does not depend on capital structure. In this case, the company cost of capital is equal to the opportunity cost of capital—that is, the cost of capital with all-equity financing.

We can turn the formula for r_A around to solve for r_E , which is the expected return to equity:

$$\begin{aligned} \text{Expected return on equity} &= \text{expected return on assets} \\ &\quad + (\text{expected return on assets} - \text{expected return on debt}) \\ &\quad \times \text{debt-equity ratio} \\ r_E &= r_A + (r_A - r_D) \frac{D}{E} \end{aligned} \tag{16.2}$$

Proposition 2

This is MM's Proposition 2: The expected rate of return on the common stock of a levered firm increases in proportion to the debt-equity ratio (D/E), expressed in market values; the rate of increase depends on the spread between r_A , the expected rate of return on a portfolio of all the firm's securities, and r_D , the expected return on the debt. Note that $r_E = r_A$ if the firm has no debt.

We can check out this formula for Macbeth Spot Removers. Before the decision to borrow

$$\begin{aligned} r_E = r_A &= \frac{\text{expected operating income}}{\text{market value of all securities}} \\ &= \frac{1,500}{10,000} = 0.15, \text{ or } 15\% \end{aligned}$$

If the firm goes ahead with its plan to borrow at $r_D = 10\%$, the expected return on assets r_A is still 15%, but the expected return on equity is

$$\begin{aligned} r_E &= r_A + (r_A - r_D) \frac{D}{E} \\ &= 0.15 + (0.15 - 0.10) \frac{5,000}{5,000} = 0.20, \text{ or } 20\% \end{aligned}$$

When the firm was unlevered, equity investors demanded a return of r_A . When the firm is levered, they require a premium of $(r_A - r_D)D/E$ to compensate for the extra risk.

MM's Proposition 1 says that financial leverage has no effect on shareholders' wealth. Proposition 2 says that the rate of return they can expect to receive on their shares increases as the firm's debt-equity ratio increases. How can shareholders be indifferent to increased leverage when it increases their expected return? The answer is that any increase in expected return is exactly offset by an increase in financial risk and therefore in shareholders' *required* rate of return.

You can see financial risk at work in our Macbeth example. Compare the risk of earnings per share in Table 16.2 versus Table 16.1. Or look at Table 16.4, which shows how a shortfall in operating income affects the payoff to the shareholders. If the firm is all-equity-financed, a decline of \$1,000 in the operating income reduces the return on the shares by 10 percentage points. If the firm issues risk-free debt with a fixed interest payment of \$500 a year, then a decline of \$1,000 in the operating income reduces the return on the shares by 20 percentage points. In other words, the effect of the proposed leverage is to double the amplitude of the swings in Macbeth's shares. Whatever the risk of the firm's shares before the refinancing, it would be twice as high afterward.

If Operating Income Falls From		\$1,500	to	\$500	Change
No debt:	Earnings per share	\$1.50		\$0.50	-\$1.00
	Return (r_E)	15%		5%	-10%
50% debt:	Earnings per share	\$2.00		0	-\$2.00
	Return (r_E)	20%		0	-20%

TABLE 16.4 Financial leverage increases the risk of Macbeth shares. A \$1,000 drop in operating income reduces earnings per share by \$1 with all-equity financing, but by \$2 with 50% debt.

Now you can see why investors require higher returns on levered equity. The required return simply rises to match the increased financial risk.

Leverage and the Cost of Equity

Consider a company with the following market-value balance sheet:

Asset value	\$100	Debt (D)	\$33.3	at $r_D = 7.25\%$
		Equity (E)	\$66.7	at $r_E = 15.5\%$
Asset value	\$100	Firm value (V)	\$100	

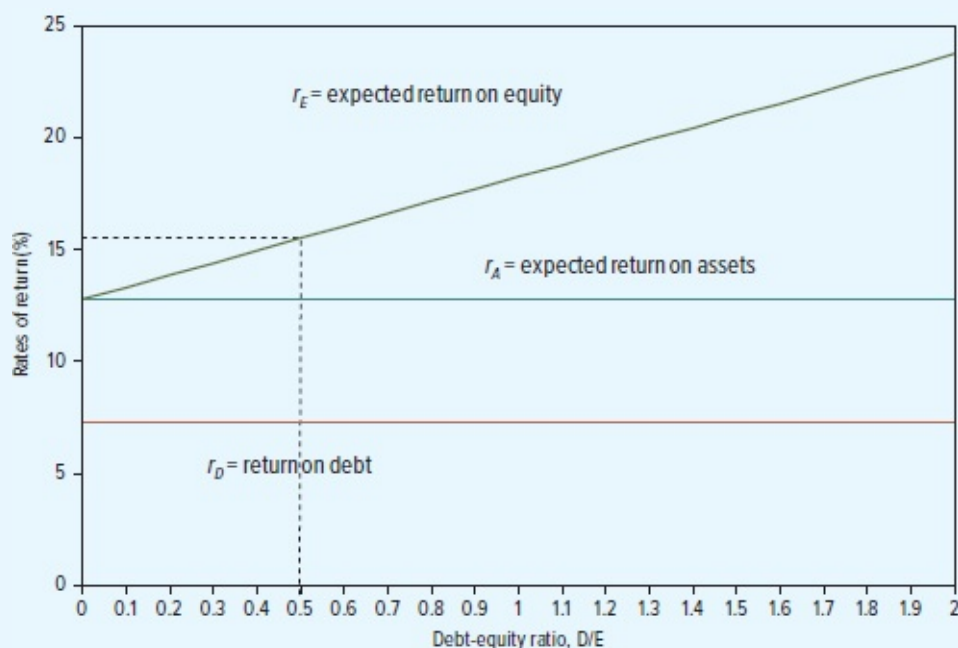
and an overall company cost of capital of

$$r_A = r_D D/V + r_E E/V$$

$$= (7.25 \times 33.3/100) + (15.5 \times 66.7/100) = 12.75\%$$

If the firm is considering a project that has the same risk as the firm's existing business, the appropriate discount rate for the cash flows is 12.75%, the firm's cost of capital.

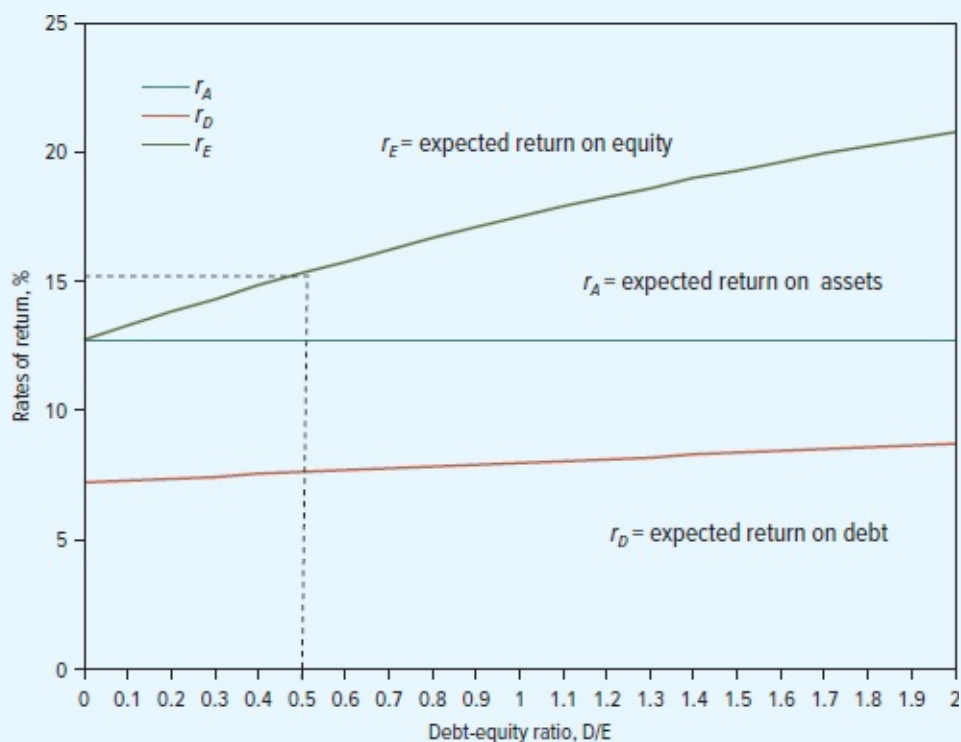
Suppose the firm changes its capital structure by issuing more debt and using the proceeds to repurchase stock. The implications of MM's Proposition 2 are shown in Figure 16.2. The required return on equity increases with the debt-equity ratio (D/E).³ Yet, no matter how much the firm borrows, the required return on the package of debt and equity, r_A , remains constant at 12.75%. How is it possible for the required return on the package to stay constant when the required return on the individual securities is changing? Answer: Because the proportions of debt and equity in the package are also changing. More debt means that the cost of equity increases but at the same time the *proportion* of equity declines.



► **FIGURE 16.2**

MM's Proposition 2 predicts that if debt is risk-free, the required return on equity, r_E , increases linearly with the debt-equity ratio, but the return on the package of debt and equity does not change.

In Figure 16.2, we have drawn the rate of interest on the debt as constant no matter how much the firm borrows. This is not wholly realistic. It is true that most large, conservative companies could borrow a little more or less without noticeably affecting the interest rate that they pay. But at higher debt levels, lenders become concerned that they may not get their money back, and they demand higher rates of interest to compensate. Figure 16.3 modifies Figure 16.2 to account for this. You can see that as the firm borrows more, the risk of the debt slowly increases. Proposition 2 continues to predict that the expected return on the package of debt and equity does not change. However, the slope of the r_E line now tapers off as D/E increases. Why? Essentially because holders of risky debt begin to bear part of the firm's operating risk. As the firm borrows more, more of that risk is transferred from stockholders to bondholders.



► **FIGURE 16.3**

If leverage increases, the risk of the debt increases and debtholders demand a higher interest rate. As lenders take on the extra risk, the expected return on equity increases more slowly. MM's Proposition 2 continues to predict that the expected return on the package of debt and equity is unchanged.

Let's assume that the firm issues an additional \$16.7 of debt and uses the cash to repurchase \$16.7 of its equity. The revised market-value balance sheet has debt of \$50 rather than \$33.3:

Asset value	\$100	Debt (D)	\$50
		Equity (E)	\$50
Asset value	\$100	Firm value (V)	\$100

The change in financial structure does not affect the amount or risk of the cash flows on the total package of debt and equity. Therefore, if investors required a return of 12.75% on the total package before the refinancing, they must require a 12.75% return on the firm's assets afterward.

Although the required return on the *package* of debt and equity is unaffected, the change in financial structure does affect the required return on the individual securities. Because the

company has more debt than before, the debtholders are likely to demand a higher interest rate. Suppose that the expected return on the debt rises to 8%. Now you can write down the basic equation for the return on assets:

$$\begin{aligned} r_A &= r_D \frac{D}{V} + r_E \frac{E}{V} \\ &= \left(8.0 \times \frac{50}{100} \right) + \left(r_E \times \frac{50}{100} \right) = 12.75\% \end{aligned}$$

Solving for the return on equity gives $r_E = 17.5\%$.

Increasing the amount of debt increased debtholder risk and led to a rise in the return that debtholders required (r_D rose from 7.25% to 8.0%). The higher leverage also made the equity riskier and increased the return that shareholders required (r_E rose from 15.5% to 17.5%). However, the weighted-average return on debt and equity was unchanged at 12.75%:

$$\begin{aligned} r_A &= r_D \frac{D}{V} + r_E \frac{E}{V} \\ &= \left(8.0 \times \frac{50}{100} \right) + \left(17.5 \times \frac{50}{100} \right) = 12.75\% \end{aligned}$$

Suppose that the company decided instead to repay all its debt and replace it with equity. In that case, all the cash flows would go to the equityholders. The company cost of capital, r_A , would stay at 12.75%, and r_E would also be 12.75%.

16.3 Self-Test

Suppose that Macbeth decides to issue \$4,000 of debt at an interest rate of 10% rather than \$5,000 as originally planned. What would happen to its cost of equity?

How Changing Capital Structure Affects the Equity Beta

We have looked at how changes in financial structure affect expected return. Let us now look at the effect on beta.

The stockholders and debtholders both receive a share of the firm's cash flows, and both bear part of the risk. For example, if the firm's assets turn out to be worthless, there will be no cash to pay to stockholders or debtholders. But debtholders usually bear much less risk than stockholders. Debt betas of large firms are typically in the range of 0 to 0.2.⁴

If you owned a portfolio of all the firm's securities, you wouldn't share the cash flows with anyone. You wouldn't share the risks with anyone either; you would bear them all. The beta of this hypothetical portfolio is just a weighted average of the debt and equity betas:

$$\beta_A = \beta_{\text{portfolio}} = \beta_D \frac{D}{V} + \beta_E \frac{E}{V} \quad (16.3)$$

Think back to our example. If the debt before the refinancing has a beta of .1 and the equity has a beta of 1.1, then

$$\beta_A = \left(0.1 \times \frac{33.3}{100}\right) + \left(1.1 \times \frac{66.7}{100}\right) = 0.767$$

What happens after the refinancing? The risk of the total package is unaffected, but both the debt and the equity are now more risky. Suppose that the debt beta changes to 0.15. We can work out the new equity beta:

$$\begin{aligned} \beta_A &= \beta_{\text{portfolio}} = \beta_D \frac{D}{V} + \beta_E \frac{E}{V} \\ 0.767 &= \left(0.15 \times \frac{50}{100}\right) + \left(\beta_E \times \frac{50}{100}\right) \end{aligned}$$

Solve for the formula for β_E . You will see that it parallels MM's Proposition 2 exactly:

$$\begin{aligned} \beta_E &= \beta_A + (\beta_A - \beta_D)D/E \\ &= 0.767 + (0.767 - 0.15)(50/50) = 1.384 \end{aligned} \quad (16.4)$$

Our example shows how borrowing creates financial leverage or gearing. Financial leverage does not affect the risk or the expected return on the firm's assets, but it does push up the risk of the common stock. Shareholders demand a correspondingly higher return because of this *financial risk*.

You can use our formulas to *unlever* betas—that is, to go from an observed β_E to β_A . You have the equity beta of 1.384. You also need the debt beta, here 0.15, and the relative market values of debt (D/V) and equity (E/V). If debt accounts for 50% of overall value V , then the unlevered beta is

$$\beta_A = \left(0.15 \times \frac{50}{100}\right) + \left(1.384 \times \frac{50}{100}\right) = 0.767$$

This runs the previous example in reverse. Just remember the basic relationship:

$$\beta_A = \beta_{\text{portfolio}} = \beta_D \frac{D}{V} + \beta_E \frac{E}{V}$$

16.4 Self-Test

Suppose that before the financing Macbeth's stock has a beta of 0.9. Macbeth now issues \$5,000 of debt with a beta of zero. What happens to

- The beta of Macbeth's assets?
- The beta of its equity?

Watch Out for Hidden Leverage

MM did not say that borrowing is a bad thing. But they insisted that financial managers stay on the lookout for the financial risk created by borrowing. That risk can be especially dangerous when the borrowing is not in plain sight. For example, most long-term leases are debt-equivalent obligations, so leases can hide debt. Long-term contracts with suppliers can also be debts in disguise when prices and quantities are fixed. For many firms pension liabilities and liabilities for employees' post-retirement health care are massive off-balance-sheet, debt-equivalent obligations.

EXAMPLE 16.1 • Reeby Sports' bocce project

Here is an example of how hidden leverage can fool a company into poor decisions. Reeby Sports is considering launch of a carbon-fiber bocce shoe. The product will require investment of \$500,000 in up-front marketing expenses and \$500,000 for new equipment. Jorge Reeby prepares a simple spreadsheet for the new product's expected five-year life and discounts at Reeby Sports' normal 10% cost of capital.⁵

Bocce Shoe Proposal						
	Cash Flow (\$1,000s)					
	0	1	2	3	4	5
Marketing	-500	+260	+260	+260	+260	+260
Equipment	-500					
Total	-1,000					
NPV at 10% =	-14, or -\$14,000					
IRR =	9.4%					

Jorge notes the negative NPV, then calls the equipment salesperson to cancel Reeby Sports' order. The salesperson, anxious to keep her sale, offers to let Reeby Sports buy the equipment now and pay later. She asks whether Jorge will commit to five fixed payments of \$122,000 per year. She argues that this will reduce the up-front investment and improve profitability. Jorge revises his spreadsheet:

Bocce Shoe Proposal (Revised)						
	Cash Flow (\$1,000s) (Fixed payments of \$122,000 per year subtracted)					
	0	1	2	3	4	5
Marketing	-500	+260	+260	+260	+260	+260
Equipment	0	-122	-122	-122	-122	-122
Total	-500	+138	+138	+138	+138	+138
NPV at 10% = +23, or \$23,000						
IRR = 11.8%						

Now Jorge is inclined to go ahead—the NPV and IRR look much better—but Jenny, his investment-banker daughter, points out that the manufacturer is really just lending \$500,000 to Reeby Sports at the same 7% interest rate that Reeby Sports would pay to a bank. She explains that the manufacturer would advance \$500,000 now in exchange for later fixed payments totaling $5 \times 122,000 = \$610,000$ undiscounted. The payments are obligatory, just like debt service on a bank loan. The effective interest rate is 7%. (You can check that the IRR to the manufacturer from agreeing to payment by installments is 7%.)

Jenny chides her father for mixing up investment and financing decisions. She upbraids him for forgetting about the financial risk created by a debt-financed equipment purchase. She berates him for discounting the cash flows of \$138,000 per year (after installment payments) at the 10% cost of capital, which is designed to value *unlevered* cash flows. “Go back to your first spreadsheet, Dad,” she instructs.⁶ Jorge, fearing chastisement, reproach, and remonstrance, agrees.

The hidden leverage in this example is, of course, only thinly disguised. The leverage would be harder to see if, for example, it were wrapped up in a financial lease transaction. See Chapter 26 and the mini-case at the end of this chapter.

16-4 No Magic in Financial Leverage

MM’s propositions boil down to the simple warning: *There is no magic in financial leverage*. Financial managers who ignore this warning can be sucked into practical mistakes. For example, suppose that someone says, “Shareholders demand—and deserve—higher expected rates of return than bondholders. Therefore, debt is the cheaper source of capital. We can reduce the company cost of capital by borrowing more.” Unfortunately, that doesn’t follow if

the extra borrowing leads shareholders to demand a still higher expected rate of return. According to MM's Proposition 2, the cost of equity capital, r_E , increases by just enough to keep the company cost of capital constant. Thus, there are actually two costs of debt. One is the interest rate that lenders require; the other is the higher return that equityholders demand to compensate them for the extra risk resulting from leverage. Mistakes arise when you ignore this second cost.

This is not the only logical short circuit that you are likely to encounter. We have cited two others in Problem 10 at the end of this chapter.

Few financial managers believe that the company cost of capital can be reduced by higher and higher leverage. But is it possible to stake out an intermediate position, in which a moderate degree of leverage increases the expected equity return, r_E , but by less than predicted by MM's Proposition 2? In this case, there would be an optimal amount of leverage that minimizes the weighted average cost of capital.

Two arguments could be advanced in support of this position. First, perhaps shareholders do not notice or appreciate the financial risk created by moderate borrowing, although they wake up when debt is "excessive." If so, shareholders in moderately leveraged firms may accept a lower rate of return than they really should.

FINANCE IN PRACTICE

Does MM Apply to Banks?

› Healthy industrial corporations typically operate with debt ratios of around 35%. Most financial managers would not be too concerned if the ratio was a few percentage points higher or lower and would probably find it difficult to put a precise figure on the optimal debt ratio. As we pointed out in Chapter 1, shareholder value is largely created by the company's choice of real assets, not by its financial structure.

This is not so for bankers. Banks operate with very high debt ratios. For example, just before the 2007–2009 financial crisis, many major banks had book debt-to-asset ratios of about 97% to 98%. So, it needed only a 2% to 3% fall in the value of their assets to wipe out the total value of the equity. With this sort of leverage, it is not surprising that banks often get into difficulties. This does not mean that banks could, or should, operate with the levels of debt that are typical in industrial companies, because a central part of their business is the issue of debt in the form of customer deposits. However, banks could reduce leverage without needing to reduce their deposits or increase their assets.

Bank regulators meeting in Basel, Switzerland, have established limits on the amount of leverage that banks should be allowed to have. Following the crisis, these limits were revised downward in the Basel III Accord. Several countries have imposed even lower limits on the amount of leverage that their banks can undertake.

These moves to make banks issue more equity capital have been vigorously opposed by bankers, who have argued that higher capital ratios would add considerably to their costs. One complaint is that a reduction in leverage would reduce their return on equity. This may be true, but it is beside the point. Increased capital would lower the expected return on equity, but MM would note that it would also reduce the risk of the equity and the return that shareholders require. In a perfect world, these two effects would cancel out so that lower leverage would not increase the cost of capital for banks and would not make shareholders any worse off.

Does bankers' opposition to higher capital requirements simply reflect a failure to understand MM's arguments, or are there other, more valid reasons for their views? One possibility is tax. Debt interest carries with it a tax shield, which may be important to a financial institution that operates on relatively fine margins. But that raises a further question: Does it make sense for the government to offer a subsidy that encourages banks to borrow if the effect of that borrowing is to cause periodic banking crises?⁷ Would it be better for the government to offer the same tax advantage to banks if they issue extra equity?

It seems unlikely that most investors are that unsophisticated.⁸ The second argument is better. It accepts MM's reasoning as applied to perfect capital markets but holds that actual markets are imperfect. Because of these imperfections, firms that borrow may provide a valuable opportunity for investors. If so, levered shares might trade at premium prices compared with their theoretical values in perfect markets.

Suppose that corporations can borrow more cheaply than investors. Then investors who want to borrow should do so indirectly by holding the stock of levered firms. They might be willing to live with expected rates of return that do not fully compensate them for the business and financial risk they bear.

Is corporate borrowing really cheaper? It's hard to say. Interest rates on home mortgages are not too different from rates on high-grade corporate bonds.⁹ Rates on margin debt (borrowing from a stockbroker with the investor's shares tendered as security) are not too different from the rates firms pay banks for short-term loans. And any investor, individual or institutional, who holds bonds can "lever up" by selling bonds and investing the proceeds in stocks. Suppose the bonds are corporate bonds. Then the rate of interest foregone on the bonds that are sold is identical to the interest rate that corporations pay on their outstanding debt.

There could nevertheless be a clientele of individual investors who would prefer corporations to borrow on their behalf. Perhaps the investors want to retain *limited liability*, which means that they are not personally liable for a corporation's debts if it defaults. Such

investors might be willing to pay a premium for levered shares *if the supply of levered shares were insufficient to meet their demands*. But all public corporations provide limited liability and most borrow. Therefore it is unlikely that financing more with debt and less with equity would induce these individual investors to pay more for *your* company.

Maybe the market for corporate leverage is like the market for automobiles. Americans need millions of automobiles and are willing to pay thousands of dollars apiece for them. But that doesn't mean that you could strike it rich by going into the automobile business. You're at least 100 years too late.

Today's Unsatisfied Clienteles Are Probably Interested in Financial Innovation

So far, we have made little progress in identifying cases where firm value might plausibly depend on financing. But our examples illustrate what smart financial managers look for. They look for an *unsatisfied* clientele, investors who want a particular kind of financial instrument but because of market imperfections can't get it or can't get it cheaply.

MM's Proposition 1 does not hold when the firm, by imaginative design of its capital structure, can offer some *financial service* that meets the needs of such a clientele. Either the service must be new and unique or the firm must find a way to provide some old service more cheaply than other firms or financial intermediaries can. In these cases, the basic requirement of Proposition 1 that investors can borrow and lend on the same terms as the firm is not satisfied and so the result does not hold.

Now, is there an unsatisfied clientele for garden-variety debt or levered equity? We doubt it. But perhaps you can invent a new security and uncover a latent demand for it.

Imperfections and Opportunities

The most serious capital market imperfections are often those created by government. An imperfection that supports a violation of MM's Proposition 1 *also* creates a money-making opportunity. Firms and intermediaries will find some way to reach the clientele of investors frustrated by the imperfection.

BEYOND THE PAGE



Bank regulation and CDOs

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For many years, the U.S. government imposed a limit on the rate of interest that could be paid on savings accounts. It did so to protect banks and savings institutions by limiting competition for their depositors' money. The fear was that depositors would run off in search of higher yields, causing a cash drain that savings institutions would not be able to meet. But

interest-rate regulation provided financial institutions with an opportunity to create value by offering money market funds. These are mutual funds that invest in Treasury bills, commercial paper, and other high-grade, short-term debt instruments. Any saver with a few thousand dollars to invest can gain access to these instruments through a money market fund and can withdraw money at any time by writing a check against his or her fund balance. Thus, the fund resembles a checking or savings account that pays close to market interest rates.¹⁰ These money market funds became enormously popular. At the peak of their popularity in 2008, they managed \$3.3 trillion of assets.

Long before all interest-rate ceilings were finally removed in 1986, most of the gains had gone out of issuing money-market funds to individual investors because the supply of them had increased to meet the demand. Once the clientele was finally satisfied, MM's Proposition 1 was restored (until the government creates a new imperfection). The moral of the story is this: If you ever find an unsatisfied clientele, do something right away, or capital markets will evolve and steal it from you.

This is actually an encouraging message for the economy as a whole. If MM are right, investors' demands for different types of securities are satisfied at minimal cost. The cost of capital will reflect only business risk. Capital will flow to companies with positive-NPV investments, regardless of the companies' capital structures. This is the efficient outcome.

16-5 A Final Word on the Cost of Capital

MM left us a simple message. When the firm changes its mix of debt and equity securities, the risk and expected returns of these securities change, but the company's overall cost of capital does not change.

Now if you think that message is too neat and simple, you're right. The complications are spelled out in the next two chapters. But we must note one complication here: In the United States and most other countries, interest paid on a firm's borrowing can be deducted from taxable income. Thus, the *after-tax* cost of debt is $r_D(1 - T_c)$, where T_c is the marginal corporate tax rate. So, when companies discount an average-risk project, they may not use the company cost of capital as we have just computed it. Instead they use the after-tax cost of debt to compute the **after-tax weighted-average cost of capital** or **WACC**:

$$\text{After-tax WACC} = r_D(1 - T_c)\frac{D}{V} + r_E\frac{E}{V}$$

The company cost of capital that we calculated in Section 16-3 is the expected return on the company's assets (r_A). It is equal to the average of the returns to the company's debt and equity investors. Unlike the company cost of capital, the after-tax WACC does *not* equal the

expected rate of return to all of the company's debt and equity investors and does not measure the opportunity cost of capital for investment in the company's assets. The after-tax WACC is lower than the opportunity cost of capital because it includes a downward tax adjustment.

EXAMPLE 16.2 • Calculating WACC for CSX

We calculated the company cost of capital for the railroad CSX in Chapter 9. The costs of debt and equity for CSX were $r_D = 3.4\%$ and $r_E = 10.3\%$. The debt-to-value ratio was 0.22. Therefore:

$$\text{Company cost of capital} = r_A = r_D D/V + r_E E/V = 8.8\%$$

Using the corporate tax rate of 21%, the after-tax cost of debt was $3.4 \times (1 - 0.21) = 2.7\%$, so WACC is:

$$\begin{aligned}\text{After-tax WACC} &= r_D(1 - T_c)\frac{D}{V} + r_E\frac{E}{V} \\ &= 3.4 \times (1 - 0.21) \times 0.22 + 10.3 \times 0.78 = 8.6\%\end{aligned}$$

MM's Proposition 2 states that *in the absence of taxes*, the company cost of capital stays the same regardless of the amount of leverage. But if companies receive a tax shield on their interest payments, then the after-tax WACC declines as debt increases. This is illustrated in Figure 16.4, which shows how CSX's WACC changes as the debt–equity ratio changes.

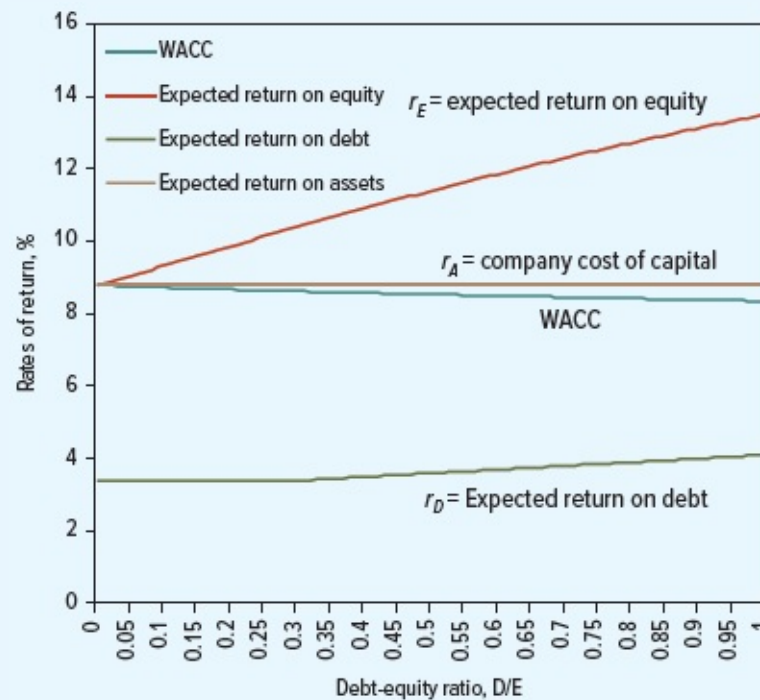


FIGURE 16.4

Estimated after-tax WACC for CSX at different debt-equity ratios. The figure assumes $r_E = 10.3\%$ at a 22% debt ratio (equivalent to a 28% debt-equity ratio) and a borrowing rate of $r_D = 3.4\%$. We assume that the debt interest rate is effectively constant at lower debt levels but increases at higher debt-equity ratios.

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Try It! Figure 16.4: Changing leverage and the cost of capital

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We have now introduced three measures of the cost of capital:

The opportunity cost of capital The opportunity cost of capital is the cost of capital for a company or a project if it is all-equity financed.

The company cost of capital (r_A) The company cost of capital is the weighted average of the costs of equity and of debt. MM tell us that in perfect markets the company cost of capital is identical to the opportunity cost for investing in the company.

The weighted average cost of capital (WACC) The weighted average cost of capital is the weighted average of the cost of equity and the *after-tax* cost of debt.

You now know how to *calculate* these measures. In Chapter 18, we will explain how to *use* them.

CSX's company cost of capital in Figure 16.4 is 8.8%. What is its opportunity cost of capital?



KEY TAKEAWAYS

- **Capital structure** A corporation's *capital structure* is its mix of debt and equity financing. A corporation that uses debt as well as equity employs *financial leverage* or *gearing*. Capital structure matters in practice, for reasons we cover in Chapters 17 and 18. But it's critical to start by understanding whether capital structure affects value in perfect financial markets, which is what MM set out to determine.
- **Modigliani and Miller** Modigliani and Miller's (MM's) Proposition 1 showed that in perfect capital markets with no taxes total market value—the value of debt and equity combined—does not depend on the proportions of debt and equity.
- **Company cost of capital** The company cost of capital, r_A , is the expected return on a portfolio of all the firm's debt and equity securities:

Company cost of capital = $r_A = r_D D/V + r_E E/V$ where r_D and r_E are the expected returns on debt and equity, D and E are the market values of debt and equity, and $V = D + E$.

Proposition 1 implies that the company cost of capital does not depend on the proportions of debt and equity.

- **Financial leverage and the cost of equity** Although the company cost of capital is not affected by leverage, the cost of equity increases as the company borrows more. MM's Proposition 2 states:

$$r_E = r_A + (r_A - r_D)D/E$$

As D/E increases, the advantage of borrowing at a cost of debt less than the cost of equity is exactly offset by the increase in the cost of equity, leaving the company cost of capital constant.

- **Financial risk** The beta of the company's assets is a weighted average of the betas of the debt and the equity:

$$\beta_A = \beta_D \frac{D}{V} + \beta_E \frac{E}{V}$$

where β_A is the beta of the firm's assets and β_D and β_E are the betas of its debt and equity. The beta of the company's assets is determined by the company's business and does not depend on the debt-equity ratio. However, the beta of the common stock does increase with the debt-equity ratio: $\beta_E = \beta_A + (\beta_A - \beta_D)D/E$

- **After-tax WACC** Do not confuse the company cost of capital with the after-tax weighted average cost of capital (WACC).

$$\text{After-tax WACC} = r_D (1 - T_C)D/V + r_E E/V$$

where T_C is the corporate tax rate. The after-tax WACC declines with financial leverage because debt interest is tax-deductible. We cover WACC in detail in Chapter 18.



FURTHER READING

The fall 1988 issue of the Journal of Economic Perspectives contains a collection of articles, including one by Modigliani and Miller, that review and assess the MM propositions. The summer 1989 issue of Financial Management contains three more articles under the heading "Reflections on the MM Propositions 30 Years Later."

Two surveys of financial innovation include:

- F. Allen and G. Yago, *Financing the Future: Market-Based Innovations for Growth*, Wharton School Publishing-Milken Institute Series on Financial Innovations (Upper Saddle River, NJ: Pearson Education, 2010).
- P. Tufano, "Financial Innovation," in G. M. Constantinides, M. Harris, and R. Stulz, eds., *Handbook of the Economics of Finance*, vol. 1A (Amsterdam: Elsevier/North-Holland, 2003), pp. 307–335.

Miller reviews the MM propositions in:

- M. H. Miller, "The Modigliani-Miller Propositions after Thirty Years," *Journal of Economic Perspectives*, 2 (Autumn 1988), pp. 99–120.

For a skeptic's view of MM's arguments see:

- S. Titman, "The Modigliani and Miller Theorem and the Integration of Financial Markets," *Financial Management* 31 (Spring 2002), pp. 101–115.



PROBLEM SETS

Please see the preface for more information.

1. **Homemade leverage (S16-2)** Sentry Insurance owns 50,000 shares of the common stock of Copperhead Corporation with a market value of \$20 per share, or \$1 million overall. Copperhead is currently financed as follows:

Market Value	
Common stock (8 million shares)	\$16 million
Short-term loans	\$ 2 million

Copperhead now announces that it is replacing \$1 million of short-term debt with an issue of common stock. What action can Sentry take to ensure that it is entitled to exactly the same proportion of profits as before?

2. **Homemade leverage (S16-2)** Companies A and B differ only in their capital structure. A is financed 30% debt and 70% equity; B is financed 10% debt and 90% equity. The debt of both companies is risk-free.
 - a. Rosencrantz owns 1% of the common stock of A. What other investment package would produce identical cash flows for Rosencrantz?
 - b. Guildenstern owns 2% of the common stock of B. What other investment package would produce identical cash flows for Guildenstern?
 - c. Show that neither Rosencrantz nor Guildenstern would invest in the common stock of B if the total value of company A were less than that of B.
3. **Corporate leverage (S16-2)** Suppose that Macbeth Spot Removers issues only \$2,500 of debt and uses the proceeds to repurchase 250 shares.
 - a. Rework Table 16-2 to show how earnings per share and share return now vary with operating income.
 - b. If the beta of Macbeth's assets is 0.8 and its debt is risk-free, what would be the beta of the equity after the debt issue?
4. **Corporate leverage (S16-2)** Reliable Gearing currently is all-equity-financed. It has 10,000 shares of equity outstanding, selling at \$100 a share. The firm is considering a capital restructuring. The low-debt plan calls for a debt issue of \$200,000 with the proceeds used to buy back stock. The high-debt plan would exchange \$400,000 of debt for equity. The debt will pay an interest rate of 10%. The firm pays no taxes.
 - a. What will be the debt-to-equity ratio if it borrows \$200,000?
 - b. If earnings before interest and tax (EBIT) are \$110,000, what will be earnings per share (EPS) if Reliable borrows \$200,000?
 - c. What will EPS be if it borrows \$400,000?
5. **MM Proposition 1 (S16-2)** Executive Chalk is financed solely by common

stock and has outstanding 25 million shares with a market price of \$10 a share. It now announces that it intends to issue \$160 million of debt and to use the proceeds to buy back common stock.

- a. How is the market price of the stock affected by the announcement?
 - b. How many shares can the company buy back with the \$160 million of new debt that it issues?
 - c. What is the market value of the firm (equity plus debt) after the change in capital structure?
 - d. What is the debt ratio after the change in structure?
 - e. Who (if anyone) gains or loses? Now try the next question.
6. **MM Proposition 1 (S16-2)** Executive Cheese has issued debt with a market value of \$100 million and has outstanding 15 million shares with a market price of \$10 a share. It now announces that it intends to issue a further \$60 million of debt and to use the proceeds to buy back common stock. Debtholders, seeing the extra risk, mark the value of the existing debt down to \$70 million.
- a. How is the market price of the stock affected by the announcement?
 - b. How many shares can the company buy back with the \$60 million of new debt that it issues?
 - c. What is the market value of the firm (equity plus debt) after the change in capital structure?
 - d. What is the debt ratio after the change in structure?
 - e. Who (if anyone) gains or loses?
7. **MM Proposition 1 (S16-2)** “MM totally ignore the fact that as you borrow more, you have to pay higher rates of interest.” Explain carefully whether this is a valid objection.
8. **MM Proposition 1 (S16-2)** Here is a limerick:
- There once was a man named Carruthers,
Who kept cows with miraculous udders.
He said, “Isn’t this neat? They give cream from one teat,
And skim milk from each of the others!”*
- What is the analogy between Mr. Carruthers’s cows and firms’ financing decisions? What would MM’s Proposition 1, suitably adapted, say about the value of Mr. Carruthers’s cows? Explain.
9. **MM’s propositions (S16-2–S16-3)** True or false?
- a. MM’s propositions assume perfect financial markets, with no distorting taxes or other imperfections.
 - b. MM’s Proposition 1 says that corporate borrowing increases earnings per share but reduces the price-earnings ratio.

- c. MM's Proposition 2 says that the cost of equity increases with borrowing and that the increase is proportional to D/V , the ratio of debt to firm value.
- d. MM's Proposition 2 assumes that increased borrowing does not affect the interest rate on the firm's debt.
- e. Borrowing does not increase financial risk and the cost of equity if there is no risk of bankruptcy.
- f. Borrowing always increases firm value if there is a clientele of investors with a reason to prefer debt.

10. MM's propositions (S16-2–S16-3) What is wrong with the following arguments?

- a. As the firm borrows more and debt becomes risky, both stock- and bondholders demand higher rates of return. Thus, by reducing the debt ratio, we can reduce both the cost of debt and the cost of equity, making everybody better off.
- b. Moderate borrowing doesn't significantly affect the probability of financial distress or bankruptcy. Consequently, moderate borrowing won't increase the expected rate of return demanded by stockholders.
- c. A capital investment opportunity offering a 10% internal rate of return is Page 481 an attractive project if it can be 100% debt-financed at an 8% interest rate.
- d. The more debt the firm issues, the higher the interest rate it must pay. That is one important reason that firms should operate at conservative debt levels.

11. MM Proposition 2 (S16-3) Spam Corp. is financed entirely by common stock and has a beta of 1.0. The firm is expected to generate a level, perpetual stream of earnings and dividends. The stock has a price-earnings ratio of 8 and a cost of equity of 12.5%. The company's stock is selling for \$50. Now the firm decides to repurchase half of its shares and substitute an equal value of debt. The debt is risk-free, with a 5% interest rate. The company is exempt from corporate income taxes. Assuming MM are correct, calculate the following items after the refinancing:

- a. The cost of equity.
- b. The overall cost of capital.
- c. The price-earnings ratio.
- d. The stock price.
- e. The stock's beta.

12. MM Proposition 2 (S16-3) "Increasing financial leverage increases both the cost of debt (r_{debt}) and the cost of equity (r_{equity}). So the overall cost of capital cannot stay constant." This problem is designed to show that the speaker is confused. Buggins Inc. is financed equally by debt and equity, each with a market value of \$1 million. The cost of debt is 5%, and the cost of equity is 10%. The company now makes a further \$250,000 issue of debt and uses the proceeds to repurchase equity. This causes the cost of debt to rise to 5.5% and the cost of equity to rise to 10.83%.

Assume the firm pays no taxes.

- a. How much debt does the company now have?
 - b. How much equity does it now have?
 - c. What is the overall cost of capital?
 - d. What is the percentage increase in earnings per share after the refinancing?
 - e. What is the new price-earnings multiple? (*Hint: Has anything happened to the stock price?*)
13. **MM Proposition 2 (S16-3)** The common stock and debt of Northern Sludge are valued at \$50 million and \$30 million, respectively. Investors currently require a 16% return on the common stock and an 8% return on the debt. If Northern Sludge issues an additional \$10 million of common stock and uses this money to retire debt, what happens to the expected return on the stock? Assume that the change in capital structure does not affect the risk of the debt and that there are no taxes.
14. **MM Proposition 2 (S16-3)** Look back to Section 16-2. Suppose that Ms. Macbeth's investment bankers have informed her that since the new issue of debt is risky, debtholders will demand a return of 12.5%, which is 2.5% above the risk-free interest rate.
- a. What are r_A and r_E ?
 - b. Suppose that the beta of the unlevered stock was 0.6. What will β_A , β_E , and β_D be after the change to the capital structure?
15. **MM Proposition 2 (S16-3)** Rodriguez Pet Foods is financed 80% by common stock and 20% by bonds. The expected return on the common stock is 12% and the rate of interest on the bonds is 6%. Assuming that the bonds are default-risk-free, draw a graph that shows the expected return of Rodriguez's common stock (r_E) and the expected return on the package of common stock and bonds (r_A) for different debt-equity ratios.
16. **MM Proposition 2 (S16-3)** Imagine a firm that is expected to produce a level stream of operating profits. As leverage is increased, what happens to
- a. The ratio of the market value of the equity to income after interest?
 - b. The ratio of the market value of the firm to income before interest if (i) MM are right and (ii) the traditionalists are right?
17. **MM Proposition 2 (S16-3)** Archimedes Levers is financed by a mixture of debt and equity. You have the following information about its cost of capital:

$r_E = \underline{\hspace{1cm}}$	$r_D = 12\%$	$r_A = \underline{\hspace{1cm}}$
$\beta_E = 1.5$	$\beta_D = \underline{\hspace{1cm}}$	$\beta_A = \underline{\hspace{1cm}}$
$r_f = 10\%$	$r_m = 18\%$	$D/V = 0.5$

Can you fill in the blanks?

- 18. MM Proposition 2 (S16-3)** Look back to Problem 17. Suppose now that Archimedes repurchases debt and issues equity so that $D/V = 0.3$. The reduced borrowing causes r_D to fall to 11%. How do the other variables change?
- 19. Debt clienteles (S16-4)** Can you invent any new kinds of debt that might be attractive to investors? Why do you think they have not been issued?
- 20. After-tax WACC (S16-5)** Gaucho Services starts life with all-equity financing and a cost of equity of 14%. Suppose it refinances to the following market-value capital structure:
- Debt (D) 45% at $r_D = 9.5\%$
- Equity (E) 55%
- a. Use MM's Proposition 2 to calculate the new cost of equity. Gaucho pays taxes at a marginal rate of $T_c = 40\%$.
 - b. Calculate Gaucho's after-tax weighted-average cost of capital.
- 21. After-tax WACC (S16-5)** Omega Corporation has 10 million shares outstanding, now trading at \$55 per share. The firm has estimated the expected rate of return to shareholders at about 12%. It has also issued long-term bonds at an interest rate of 7% and has a debt value of \$200 million. It pays tax at a marginal rate of 21%.
- a. What is Omega's after-tax WACC?
 - b. How much higher would WACC be if Omega used no debt at all? (*Hint:* For this problem, you can assume that the firm's overall beta [β_A] is not affected by its capital structure or by the taxes saved because debt interest is tax-deductible.)
- 22. After-tax WACC (S16-5)** Gamma Airlines has an asset beta of 1.5. The risk-free interest rate is 6%, and the market risk premium is 8%. Assume the capital asset pricing model is correct. Gamma pays taxes at a marginal rate of 25%. Draw a graph plotting Gamma's cost of equity and after-tax WACC as a function of its debt-to-equity ratio D/E , from no debt to $D/E = 1.0$. Assume that Gamma's debt is risk-free up to $D/E = 0.25$. Then the interest rate increases to 6.5% at $D/E = 0.5$, 7% at $D/E = 0.8$, and 8% at $D/E = 1.0$. As in Problem 21, you can assume that the firm's overall beta (β_A) is not affected by its capital structure or the taxes saved because debt interest is tax-deductible.

CHALLENGE

- 23. Investor choice (S16-4)** Consider the following three tickets: Ticket A pays \$10 if is _____ elected as president, ticket B pays \$10 if is _____ elected, and ticket C pays \$10 if neither is elected. (Fill in the blanks yourself.) Could the three tickets sell for less than the present value of \$10? Could they sell for more? Try auctioning off the tickets. What are the implications for MM's Proposition 1?

- 24. Investor choice (S16-4)** People often convey the idea behind MM's Proposition 1 by various supermarket analogies, for example, "The value of a pie should not depend on how it is sliced," or, "The cost of a whole chicken should equal the cost of assembling one by buying two drumsticks, two wings, two breasts, and so on."

Actually Proposition 1 doesn't work in the supermarket. You'll pay less for an uncut whole pie than for a pie assembled from pieces purchased separately. Supermarkets charge more for chickens after they are cut up. Why? What costs or imperfections cause Proposition 1 to fail in the supermarket? Are these costs or imperfections likely to be important for corporations issuing securities on the U.S. or world capital markets? Explain.

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- 25. Investor choice (S16-4)** Suppose that new security designs could be patented.¹¹ The patent holder could restrict use of the new design or charge other firms royalties for using it. What effect would such patents have on MM's capital-structure irrelevance theory?



SOLUTIONS TO SELF-TEST QUESTIONS

- 16.1** The value of the equity increases by \$5 million as long as debtholders can demand only a fair (zero-NPV) rate of interest (and the value of any existing debt is unaffected).
- 16.2** Debt interest is $0.10 \times 50 = \$5$ million (40% of earnings). The remaining \$7.5 million (or 60% of earnings) goes to the equity holders. The value of the debt is $5/0.1 = \$50$ million or half of the company's value. Debt investors put up half the capital, but because their investment is safer, they are entitled to less than half the firm's income.
- 16.3**
$$r_E = r_A + (r_A - r_D)D/E$$
$$= 15 + (15 - 10) \times (4/10) = 17\%$$
- 16.4**
- a. The beta of the assets does not change.
- $$\beta_E = \beta_A + (\beta_A - \beta_D)D/E$$
- b.
$$= 0.9 + (0.9 - 0) \times (50/50) = 1.8$$
- 16.5** Also 8.8%. MM would say that in the absence of taxes, the company cost of capital is the same when there is no debt.

Claxton Drywall Comes to the Rescue

A law firm (not Dewey, Cheatem, and Howe) is expanding rapidly and must move to new office space. Business is good, and the firm is encouraged to purchase an entire building for \$10 million. The building offers first-class office space, is conveniently located near their most important corporate clients, and provides space for future expansion. The firm is considering how to pay for it.

Claxton Drywall, a consultant, encourages the firm not to buy the building but to sign a long-term lease for the building instead. “With lease financing, you’ll save \$10 million. You won’t have to put up any equity investment,” Drywall explains.

The senior law partner asks about the terms of the lease. “I’ve taken the liberty to check,” Drywall says. “The lease will provide 100% financing. It will commit you to 20 fixed annual payments of \$950,000, with the first payment due immediately.”

“The initial payment of \$950,000 sounds like a down payment to me,” the senior partner observes sourly.

“Good point,” Drywall says amiably, “but you’ll still save \$9,050,000 up front. You can earn a handsome rate of return on that money. For example, I understand you are considering branch offices in London and New Delhi. The \$9 million would pay the costs of setting up the new offices, and the cash flows from the new offices should more than cover the lease payments. And there’s no financial risk—the cash flows from the expansion will cover the lease payments with a safety cushion. There’s no reason for you or your partners to worry or to demand a higher-than-normal rate of return.”

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QUESTIONS

Suppose the present value of the building equals its purchase price of \$10 million. Assume that the law firm can finance the offices in London and New Delhi from operating cash flow, with cash left over for the lease payments. The firm will not default on the lease payments. For simplicity you can ignore taxes.

1. If the law firm takes the lease, it will invest \$950,000 and, in effect, borrow \$9,050,000, repaid by 19 installments of \$950,000. What is the interest rate on this disguised loan?
2. The law firm could finance 80% of the purchase price with a conventional mortgage at a 7% interest rate. Is the conventional mortgage better than the lease?
3. Construct a simple numerical example to convince Drywall that the lease would expose the law firm to financial risk. [Hint: What is the rate of return on the firm’s equity investment in the office building if a recession arrives and the market value of the (leased) office building falls to \$9 million after one year? What is the rate of return with conventional mortgage financing? With all-equity financing?]

4. Do the investments in London and New Delhi have anything to do with the decision to finance the office building? Explain briefly.

¹F. Modigliani and M. H. Miller, “The Cost of Capital, Corporation Finance and the Theory of Investment,” *American Economic Review* 48 (June 1958), pp. 261–297. MM’s basic argument was anticipated in 1938 by J. B. Williams and to some extent by David Durand. See J. B. Williams, *The Theory of Investment Value* (Cambridge, MA: Harvard University Press, 1938); and D. Durand, “Cost of Debt and Equity Funds for Business: Trends and Problems of Measurement,” *Conference on Research in Business Finance* (New York: National Bureau of Economic Research, 1952, pp. 215–262).

²Rather than borrow on your own account, you might be able to lend $0.01 D_L$ less than you commonly do. The effect is the same.

³Note that the firm’s debt ratio (D/V) of 0.333 corresponds to a debt-equity ratio (D/E) of $0.333/0.667 = 0.5$. Figure 16.2 shows that the required return on equity is 15.5% when the debt-equity ratio = 0.5.

⁴Debt betas are often close to zero but can move into positive territory for two reasons. First, if the risk of default increases, more of the firm’s business risk is shifted to lenders. Thus, “junk” debt issues typically have positive betas. Second, changes in interest rates can affect both stock and bond prices, creating a positive correlation between returns on bonds and returns on the stock market. This second reason is most important when long-term interest rates are unusually volatile, as in the United States in the 1970s and early 1980s.

⁵Reeby Sports has massive tax losses carried forward from the disastrous recession of 2025. Therefore, Jorge’s cash-flow projections assume no taxes and ignore depreciation tax shields.

⁶Jorge might try discounting the cash flows in his second spreadsheet at a cost of *equity*. We discuss the “flow to equity” valuation method in Chapter 18. This method mixes investment and financing decisions, however, and is rarely used to value individual projects. Jorge is well-advised to calculate NPV from his first spreadsheet and *then* ask whether the installment sale adds value, compared with other sources of financing.

⁷For a discussion of these issues by four proponents of higher bank capital requirements, see A. R. Admati, P. M. DeMarzo, M. F. Hellwig, and P. Pfleiderer, “Fallacies and Irrelevant Facts in the Debate on Capital Regulation” in C. Goodhart, D. Gabor, J. Vestergaard and I. Erturk, eds., *Central Banks at a Crossroads: Europe and Beyond* (London: Anthem Press, 2014).

⁸This first argument may reflect a confusion between financial risk and the risk of default. Default is not a serious threat when borrowing is moderate; stockholders worry about it only when the firm goes “too far.” But as our Macbeth example in Section 16-2 illustrated, stockholders bear financial risk—in the form of increased volatility of rates of return and a higher beta—even when the chance of default is nil.

⁹One of the authors once obtained a home mortgage at a rate $1/2$ percentage point *less* than the contemporaneous yield on long-term AAA bonds.

¹⁰Money market funds are not totally safe. In 2008, the Reserve Primary Fund incurred heavy losses on its holdings of Lehman Brothers debt and became only the second money market fund in history to “break the buck” by paying investors only 97 cents on the dollar. Since then, additional regulations have been introduced to prevent a repetition of this failure.

¹¹So far, security designs cannot be patented, but other financial applications have received patent protection. See J. Lerner, “Where Does *State Street* Lead? A First Look at Finance Patents, 1971–2000,” *Journal of Finance* 57 (April 2002), pp. 901–930.